

SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Short Answer Test

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems. (BTL 3)

Assessment Tool Weightage: 40%

QUESTIONS: 10

Total Marks: 50

Annexure A Short Answer Test

Code of Conduct:

I P. Haritha (192425254) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P. Haritha

89
105

Short Answer Test Questions

1. A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.
2. Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.
3. An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.

4. Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.
 5. A surveillance system requires real-time image processing. Classify the level of Computer Vision involved and justify your choice.
 6. Explain how image formation models help in improving image quality in applications such as autonomous driving.
 7. A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.
 8. Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.
 9. In an application like facial recognition, explain how different levels of Computer Vision are involved from image acquisition to decision making.
 10. An image captured with low resolution fails to detect objects clearly. Explain how sampling and image formation affect object visibility.
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Assessment - 1

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Short Answer Test

ITA 0509

- 1, A camera captures an image that appears blurred due to improper sampling. Explain how sampling affects image quality and suggest a method to correct it.

Sampling is the process of converting a continuous image into a discrete digital image by dividing it into pixels. The sampling rate determines the number of pixels used to represent the image. If the sampling rate is too low, important details are lost, resulting in a blurred image and aliasing effects. High sampling preserves fine details and improves image sharpness. To correct improper sampling, a higher resolution image sensor or camera should be used. Anti-aliasing filters should also be applied before sampling to remove high-frequency components. Proper interpolation techniques such as bilinear or bicubic interpolation can further improve image quality. Therefore, appropriate sampling is essential for accurate image representation.

- 2, Given a real-world scene captured under low light, describe how the image sensing and acquisition process influences the final digital image.

Image acquisition begins when light reflected from a real-world object passes through the camera

lens and reaches the image sensor (CCD or CMOS) in low-light conditions, the sensor receives insufficient light, producing dark images with reduced contrast and increased electronic noise. This affects brightness, color accuracy, and image clarity. Sensor sensitivity, increasing exposure time, improving illumination, or using high-sensitivity sensors can significantly enhance the captured image. A noise reduction system can also improve image quality. Hence, the image acquisition process directly determines the quality and usability of the digital image.

3. An image shows aliasing artifacts. Identify the cause using sampling and quantization concepts and propose a corrective approach.

Aliasing is a distortion that occurs when an image is sampled below the Nyquist sampling rate. As a result, fine details are incorrectly represented as jagged edges or false patterns. Improper quantization with fewer intensity levels can further reduce image quality by producing visible intensity changes. These effects decrease image accuracy and clarity. Aliasing can be minimized by increasing the sampling frequency and applying an anti-aliasing (low-pass) filter before sampling. Using higher-bit quantization is essential for producing high-quality digital images.

Differentiate how pixel resolution and intensity resolution affect image representation in a practical application like medical imaging.

pixel resolution refers to the number of pixels used to represent an image and determines its spatial detail. higher pixel

→ Intensity Resolution: Determines the number of gray levels. higher intensity resolution allows better differentiation between tissues.

In medical hanging both are essential for accurate of diagnosis.

5. A Surveillance System requires real-time image processing. Classify the level of Computer vision involved and Justify your choice.

This System mainly uses high-level computer vision because it not only captures images but also detects objects recognizes face, tracks movements, and make decisions instantly. Real-time processing enables immediate responses for security monitoring.

6. Explain how image formation models helps in improving image quality in applications such as autonomous driving.

Image formation models describe how light interacts with the objects and the camera. they help correct lighting variations, lens distortion, shadows

and reflections

→ this improves image quality, allowing autonomous vehicles to accurately detect lanes, traffic signs, pedestrians and obstacles.

7) A grayscale image is converted into a digital image describe the role of quantization in this process and its impact on image detail.

→ quantization converts continuous grayscale values into discrete intensity levels

→ More quantization levels preserve image details and smooth transitions, while fewer levels cause loss of detail and produce banding effects. therefore higher quantization improves image quality.

8) Given a noisy image acquired from a sensor, explain how the acquisition process contributes to noise and how it can be minimized.

Noise is introduced during image acquisition due to low light, sensor limitations, electronic interference and high temperature.

→ Noise can be minimized by using high-quality sensors, proper lighting, cooling mechanisms, longer exposure and image filtering techniques such as median or gaussian filtering.

9) In an application like facial recognition, explain how different levels of computer vision are involved from image acquisition to decision making.

Low-level vision: Captures the image and removes noise

Mid-level vision

Detects facial features such as eyes, nose and mouth.

High-level vision

Recognizes the person's identity and make a decision such as granting or denying access.

All three levels work together for accurate facial recognition.

2) An image captured with low resolution fails to predict objects clearly. Explain how sampling and image formation affect object visibility.

→ Low sampling produces fewer pixels, reducing image details and making objects difficult to identify.

→ poor image formation due to incorrect focus, lighting or lens distortion further reduces visibility

→ higher sampling resolution and improved image formation produce sharper images and better object detection.